Vikas Reddy Meka

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Education

- **Indian Institute of Technology Patna** — Patna, India
  B.Tech – Computer Science and Engineering — *Aug 2024 – May 2028*
  - **Relevant Coursework**: Data Structures, Algorithms, Graph Theory, Optimization, Machine Learning
- **Sri Guru Junior College** — Hyderabad, India
  Intermediate – MPC (93.7%)
  - **Achievement**: AIR 2996 in JEE Advanced 2024

Technical Skills

- **Languages**: C++, C, Python
- **Data Structures**: Arrays, Trees, Heaps, Graphs
- **Algorithms**: Dijkstra, BFS, DFS, Greedy, Dynamic Programming
- **Machine Learning**: Logistic Regression, Random Forest, Model Evaluation, Cross-Validation
- **Tools**: Linux, Git, VS Code, LaTeX

Academic & Technical Experience

- **Emergency Facility Coverage and Routing System**
  Graph Algorithms Project
  - **System Design**: Modeled a city-scale road network with **1,000+ nodes** using weighted graphs
  - **Algorithm Implementation**: Implemented multi-source Dijkstra with complexity **O((V+E) log V)**
  - **Analysis**: Identified **15–20% underserved regions** using response-time thresholds
  - **Optimization**: Proposed facility placements reducing worst-case response delay by **30%**
  - **Technologies**: C++, STL, Graph Algorithms
- **Student Performance and Dropout Risk Prediction**
  Machine Learning Project
  - **Data Processing**: Analyzed academic and attendance data for **100+ students**
  - **Modeling**: Built Logistic Regression and Random Forest classifiers
  - **Evaluation**: Achieved **93% accuracy** and **ROC-AUC of 0.94**
  - **Interpretability**: Performed feature importance analysis to explain predictions
  - **Technologies**: Python, Pandas, NumPy, Scikit-learn
- **Customer Churn Prediction using Behavioral Data**
  Machine Learning Project
  - **Problem Definition**: Predicted customer churn for **5,000+ users** using behavioral data
  - **Modeling**: Trained Logistic Regression and Random Forest classifiers
  - **Evaluation**: Achieved **ROC-AUC of 0.88+** with stratified cross-validation
  - **Business Insight**: Identified high-risk segments to support retention strategies
  - **Technologies**: Python, Pandas, NumPy, Scikit-learn

Competitive Programming

- **Codeforces**: Specialist (1470), solved **200+** problems
- **Focus Areas**: Graphs, Greedy, Binary Search, Dynamic Programming

Leadership

- **Class Representative** — Aug 2024 – Aug 2025
  Indian Institute of Technology Patna
  - **Representation**: Acted as the official liaison between **faculty and 80+ students** for academic matters
  - **Coordination**: Organized communication regarding schedules, evaluations, and academic announcements
  - **Problem Resolution**: Collected and presented student concerns to faculty, helping resolve issues efficiently
  - **Leadership**: Led discussions and facilitated consensus among peers on academic decisions